Database Systems

Entity Relationship Model

Entity Relationship Model

✓**The Entity Relationship Model**

- Database design
- Entity, Entity Set, Attribute, Relationship

- E/R Design Considerations
  - Constraints: Key, Referential, Degree
  - Relationship Conditions: Multiplicity of Relationships, Multiway Relationships

- More E/R Concepts
  - Combining Relations, Constraints, Subclasses, Weak Entity Sets

- Conversion to SQL

Database Design Process

1. Requirements Analysis → 2. Conceptual Design → 3. Logical, Physical, Security, etc.

Requirements Analysis

- What data is going to be stored?
- What are we going to do with the data?
- Who should access the data?

Database Design Process

1. Requirements Analysis → 2. Conceptual Design → 3. Logical, Physical, Security, etc.

Conceptual Design

- A high-level description of the database
- Sufficiently precise that technical people can understand it
- But not so precise that non-technical people can not participate

✓**Where E/R Model Fits**

Database Design Process

1. Requirements Analysis → 2. Conceptual Design → 3. Logical, Physical, Security, etc.

Detailed Design

- A Logical Database Design
- Physical Database Design
- Security Design

Entity Relationship Model

- Principal element types

  - **Entity:** Individual object

  - **Entity Sets**: Collection of similar entities forms an entity set
    ER model is a static concept, involving the structure of data and **not the operations on data**

  - **Attributes**: Entity sets have attributes, showing properties of the entities in that set

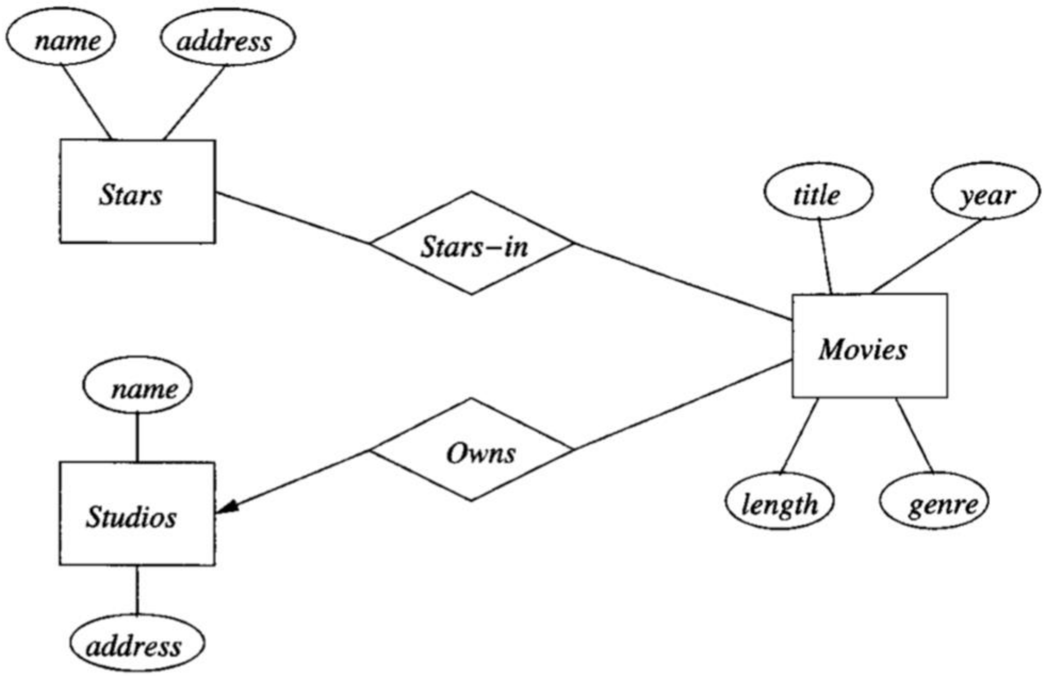
  - **Relationships**: Connections among two or more entity sets

Entity Relationship Model

- **Example:** Movie Database
  - Each movie is an entity, and the set of all movies constitutes an entity set.
  - The stars are entities, and the set of stars is an entity set
  - A studio is another kind of entity, and the set of studios is a third entity set
  - The entity set Movies might be given attributes such as title and length
  - Movies and Stars are two entity sets; we could have a relationship Stars-in that connects movies and stars
  - Entities are **not** explicitly represented in E/R diagrams!

ER Diagrams

- A graph representing entity sets, attributes, and relationships
  - **Entity sets** are represented by **rectangles**.
  - **Attributes** are represented by **ovals**.
  - **Relationships** are represented by **diamonds**.



From Chapter 4, The complete book

Keys

- A **key** is a set of attributes that uniquely identifies an entity.

- Every entity must have a key, although in some cases (we will see later) the key actually belongs to another entity set.

- Denote elements of the primary key by <u>underlining</u>.

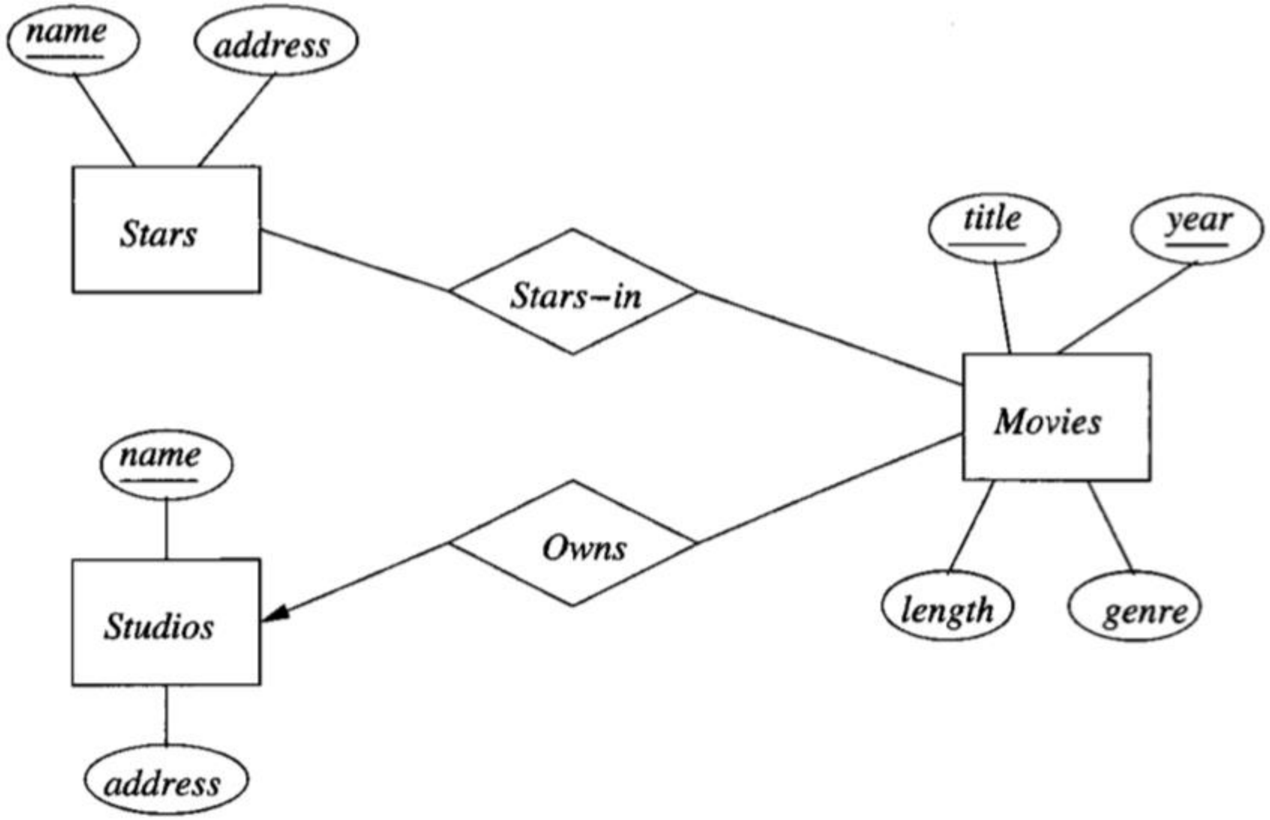
- There is no notation for representing the situation where there are several keys for an entity set.

- We underline only the primary key.

- Multiple underlining: they are each member of the key.

Keys

- The attributes title and year together form the key for **Movies**.



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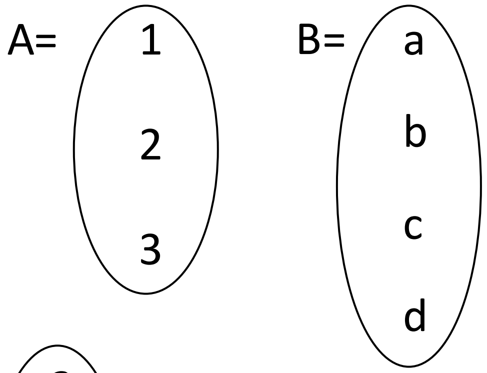
What is a Relationship?

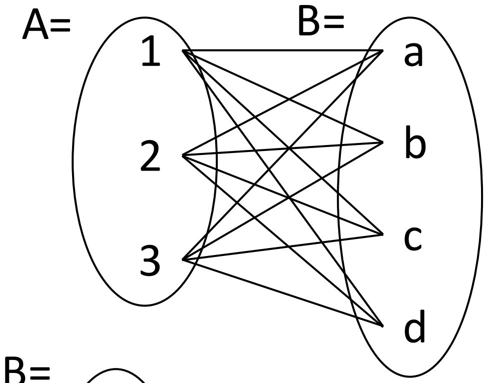
- Let **A** and **B** be sets
  - `A={1,2,3}, B={a,b,c,d}`
- A x B (the **cross-product**) is the set of all pairs **`(a,b)`**
  - `A × B = {(1,a), (1,b), (1,c), (1,d), (2,a), (2,b), (2,c), (2,d), (3,a), (3,b), (3,c), (3,d)}`
- We define a **relationship** to be a **subset of A x B**
  - `R = {(1,a), (2,c), (2,d), (3,b)}`

What is a Relationship?

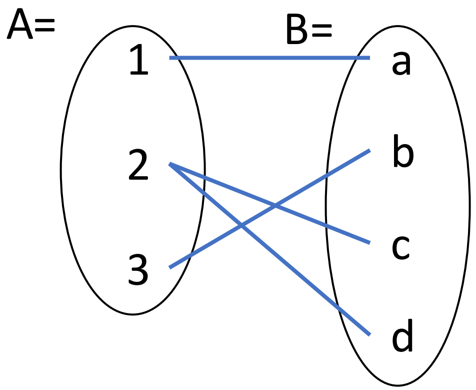
- `A={1,2,3}`
  `B={a,b,c,d}`



- $A \times B = \{(1,a),\ (1,b),\ (1,c),\ (1,d),\ (2,a),\ (2,b),\ (2,c),\ (2,d),\ (3,a),\ (3,b),\ (3,c),\ (3,d)\}$



- $R = \{(1,a),\ (2,c),\ (2,d),\ (3,b)\}$



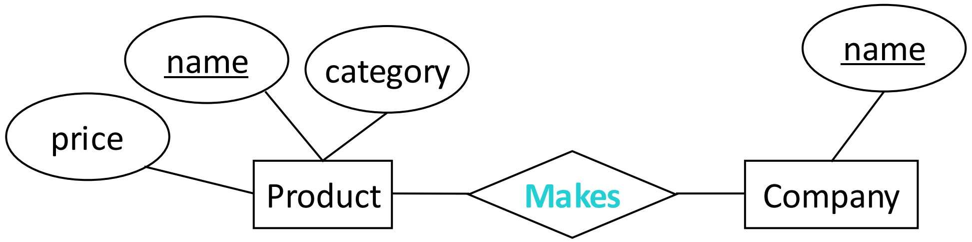
What is a Relationship?

Company

<u>name</u>
Apple
Microsoft

Product

<u>name</u>	category	price
iPhone 8	Electronics	$700
iPad 4	Electronics	$300
Office	Software	$120



A **relationship** between **entity sets P and C** is a **subset of all possible pairs of entities in P and C**, with tuples uniquely identified by **P and C's keys.**

What is a Relationship?

Company

name
Apple
Microsoft

Product

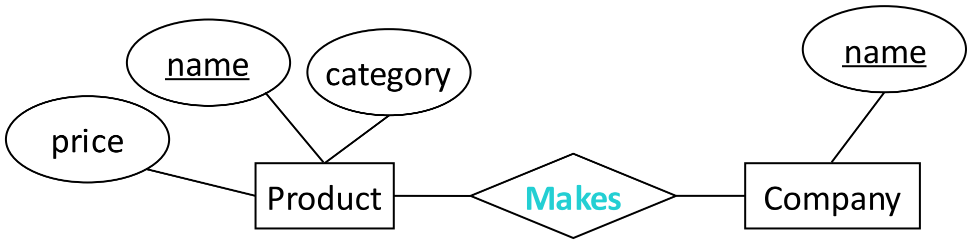
name	category	price
iPhone 8	Electronics	$700
iPad 4	Electronics	$300
Office	Software	$120

Company C × Product P

C.name	P.name	P.category	P.price
Apple	iPhone 8	Electronics	$700
Apple	iPad 4	Electronics	$300
Apple	Office	Software	$120
Microsoft	iPhone 8	Electronics	$700
Microsoft	iPad 4	Electronics	$300
Microsoft	Office	Software	$120

Makes

C.name	P.name
Apple	iPhone 8
Apple	iPad 4
Microsoft	Office



What is a Relationship?

- There can only be **one relationship for every unique combination of entities**

- This also means that **the relationship is uniquely determined by the keys of its entities**

- Example: the key for Makes is

  {Product.name, Company.name}

Makes

C.name	P.name
Apple	iPhone 8
Apple	iPad 4
Microsoft	Office

Entity Relationship Model

- The Entity Relationship Model
  - Database design
  - Entity, Entity Set, Attribute, Relationship, Key

✓ **E/R Design Considerations**
  - Constraints: Referential Constraints, Degree Constraints
  - Relationship Conditions: Multiplicity of Relationships, Multiway Relationships

- More E/R Concepts
  - Combining Relations, Constraints, Subclass, Weak Entity Sets

- Conversion to SQL

Referential Integrity

- A value appearing in one context must also appear in another

- A rounded arrow head pointing to *F*
  - Not only the relationship is many-one from E to F
  - But also the entity of set F related to a given entity of set E is required to exist

E/R diagram showing referential integrity constraints

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Degree Constraints

- A bounding number on the edges that connect a relationship to an entity set

- Indicates limits on the number of entities that can be connected to any one entity of the related entity set

Representing a constraint on the number of stars per movie

From Chapter 4, The complete book

Multiplicity of Binary ER Relationships

- If each member of **E** can be connected by **R** to at most one member of **F**: **R is many-one from E to F**.
- If **R** is both many-one from **E** to **F** and many-one from **F** to **E**: **R is one-one**.
- If **R** is neither many-one from **E** to **F** nor from **F** to **E**: **R is many-many**.

A one-one relationship

From Chapter 4, The complete book

Multiplicity of ER Relations

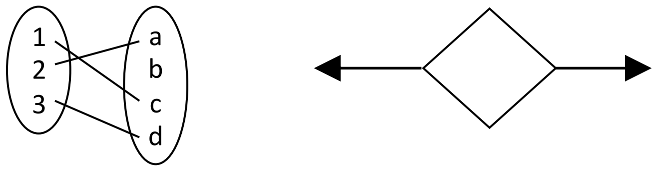
- Usage of arrows

  X → Y means

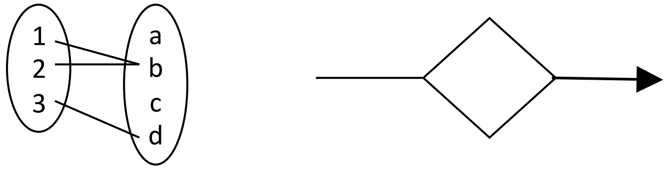
  There exists a function mapping from X to Y

  (Recall the definition of a function)

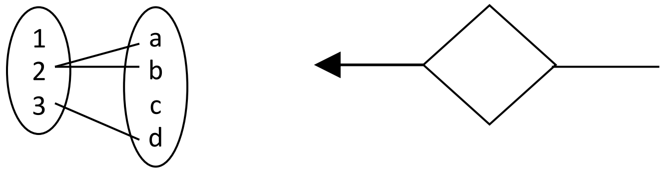
One-to-one:



Many-to-one:



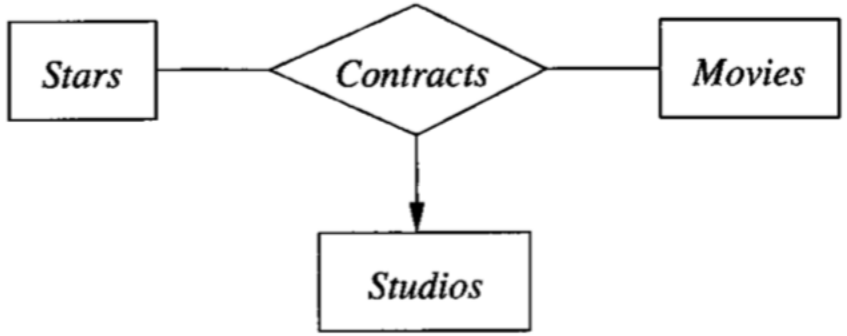
One-to-many:



Many-to-many:

Multiway Relationships

- Relationships involving more than two entity sets
- A multiway relationship in an E/R diagram is represented by lines from the relationship diamond to each of the involved entity sets.



A ternary (three-way) relationship

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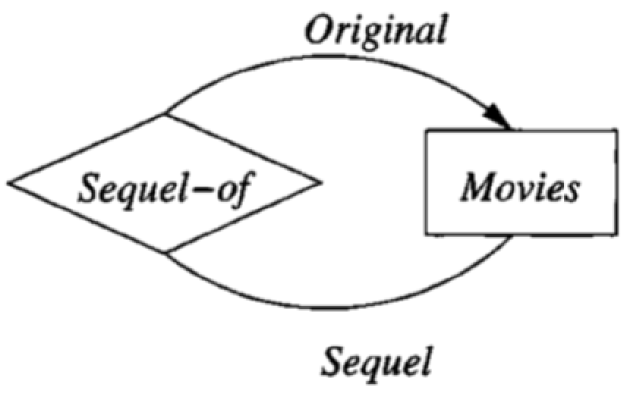
Roles in Relationships

- It is possible that one entity set appears two or more times in a single relationship.
- Each line to the entity set represents a different **role** that the entity set plays in the relationship.
- We label the edges between the entity set and relationship by names (roles).



A relationship with roles

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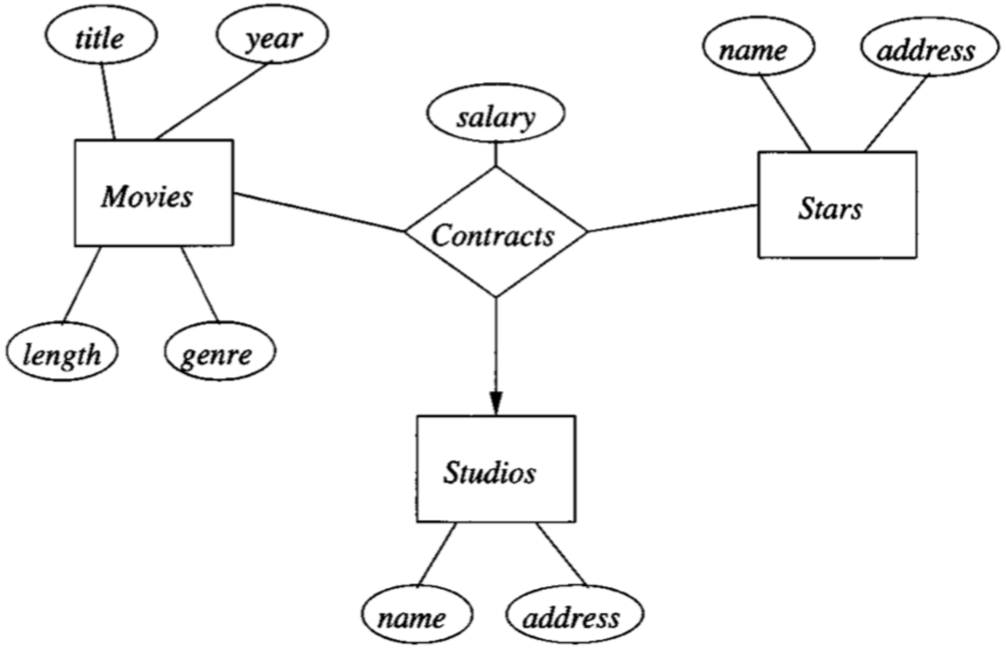
Attributes in Relationships

- It is never necessary to place attributes on relationships.
- Could be replaced by a new entity set with the attributes.



A relationship with an attribute

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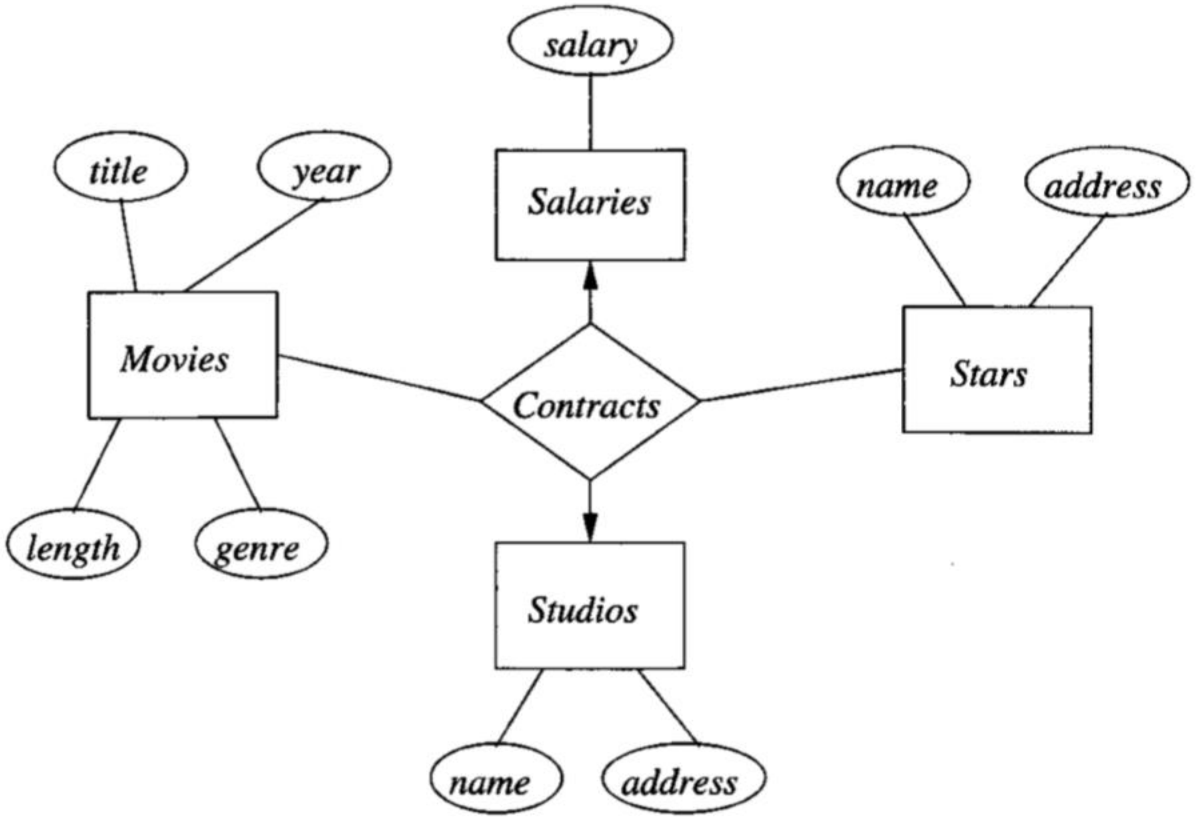
Attributes in Relationships



Moving the attribute to an entity set

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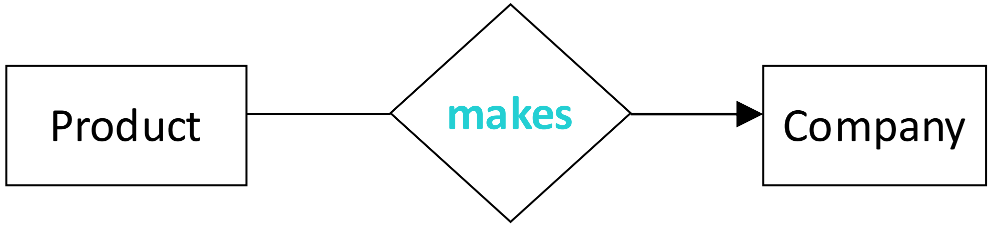
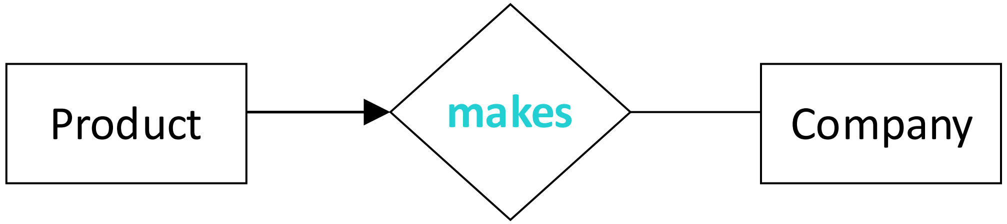
A Note on Different Notations



Each product made by at most one company.



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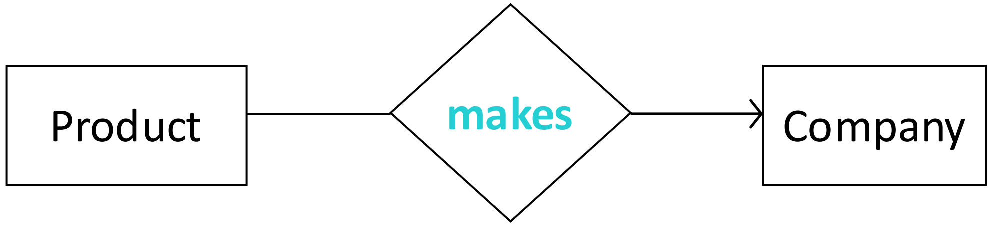
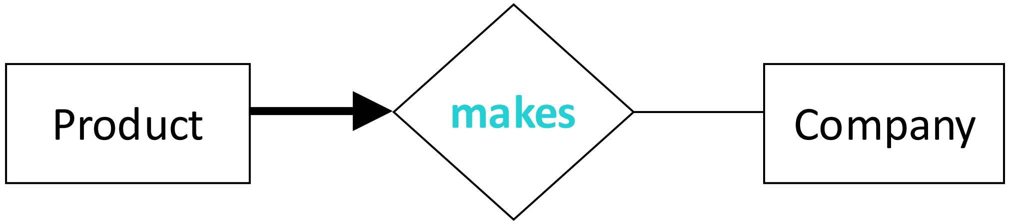


Referential Integrity.



Each product made by *exactly* one company.

Entity Relationship Model

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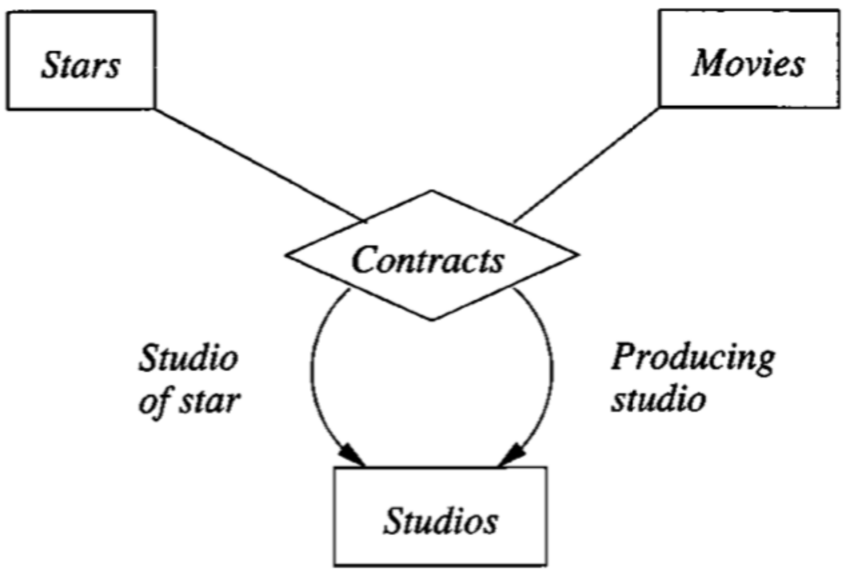
  - Database design
  - Entity, Entity Set, Attribute, Relationship
- E/R Design Considerations
  - Constraints: Key, Referential, Degree
  - Relationship conditions: Multiplicity of Relationships, Multiway Relationships
- ✓ **More E/R Concepts**
  - Constraints, Subclasses, Weak Entity Sets
- Conversion to SQL

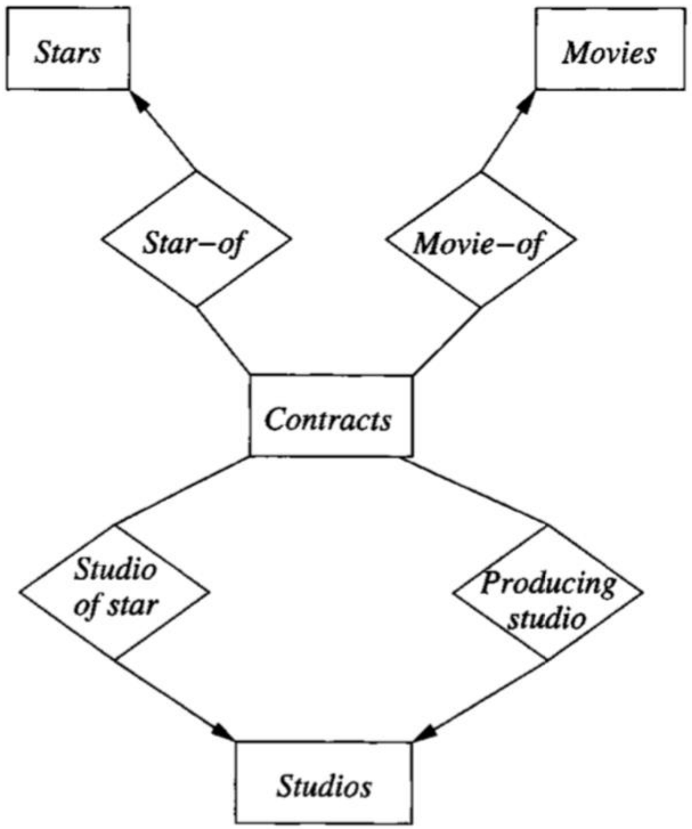
Multiway to Binary

- E/R Model does not require binary relationships

- Introduce a new entity set: a **connecting entity set**
  - Entities as tuples of the relationship set for the multiway relationship
  - Many-one relationships from the connecting entity set to each of the entity sets that provide components of tuples in the original, multiway relationship
  - If an entity set plays more than one role, then it is the target of one relationship for each role

Multiway to Binary





Replacing a multiway relationship by an entity set and binary relationships

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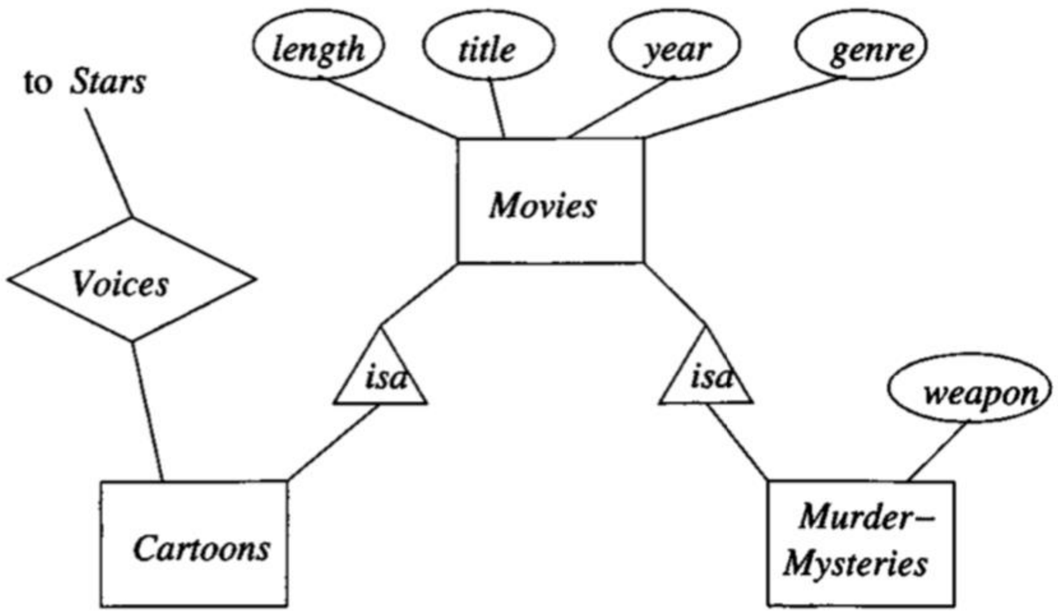
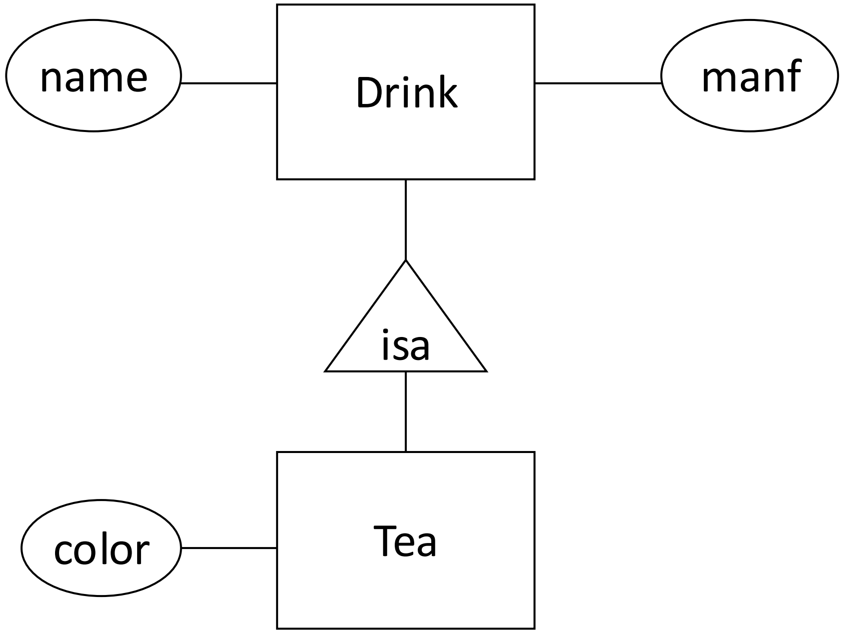
Subclasses

- When entity set has entities that have special properties not associated with all members of the set
  - Special-case entity sets: subclasses
  - Expressed as a special kind of relationship: **isa** relationship
  - Notation: **triangle**
  - Every **isa** relationship is **one-one**

Subclasses





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Weak Entity Sets

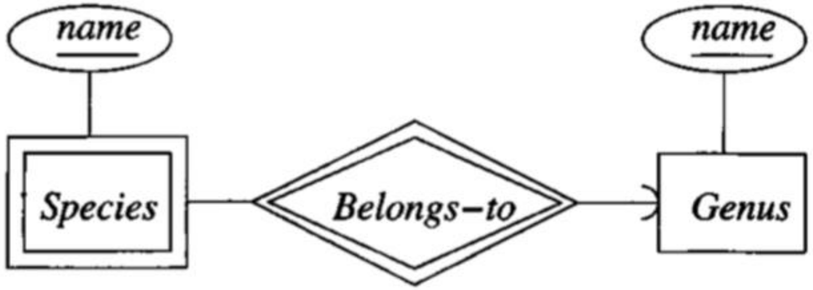
- When the key of an entity set is composed of attributes, some or all of which belong to another entity set

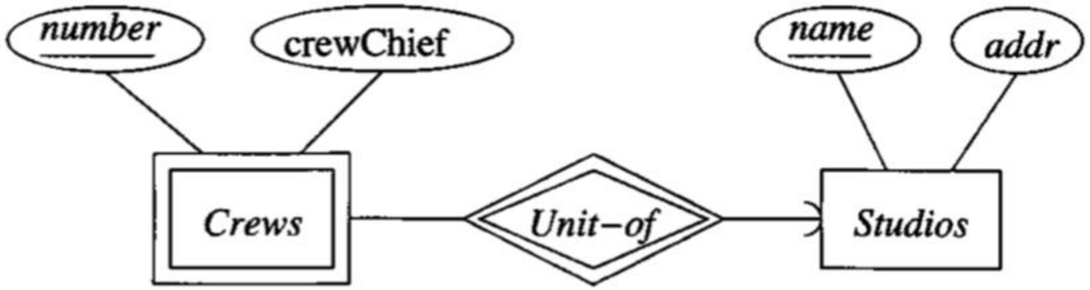
- **Causes**
  - Falling into a hierarchy based on classifications unrelated to the isa-hierarchy
  - Connecting entity sets as a way to eliminate a multiway relationship

Weak Entity Sets

- The double-rectangle indicates a weak entity set
- The double-diamond indicates a many-one relationship that helps provide the key for the weak entity set

- Examples





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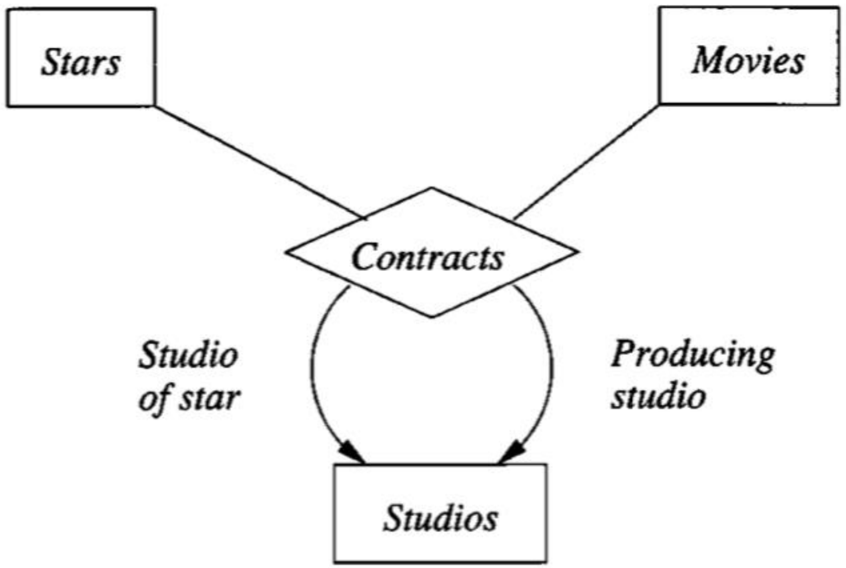
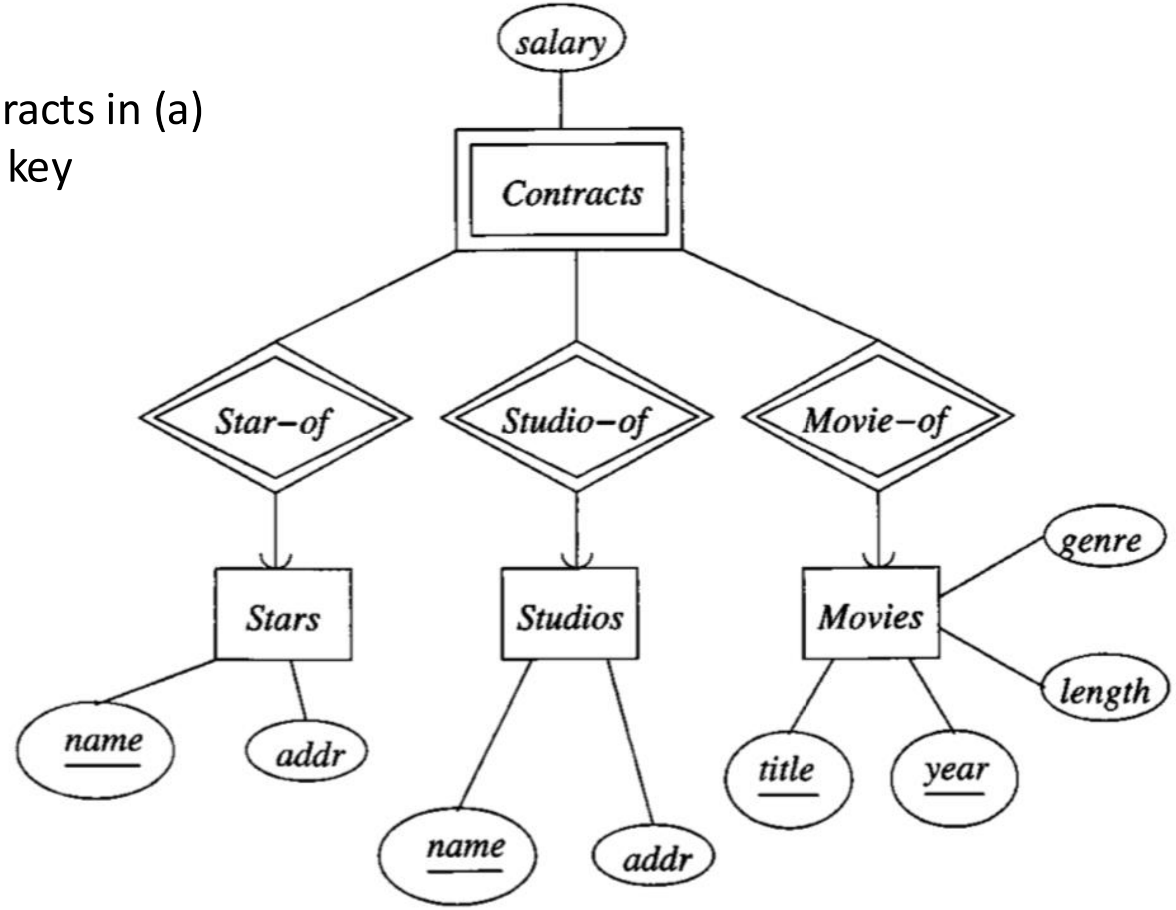
Weak Entity Sets

Contracts in (b) replaces the ternary relationship Contracts in (a)
Contracts has salary, but it does not contribute to the key



(a)



(b)

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- More E/R Concepts
  - Combining, Constraints, Subclasses, Weak Entity Sets

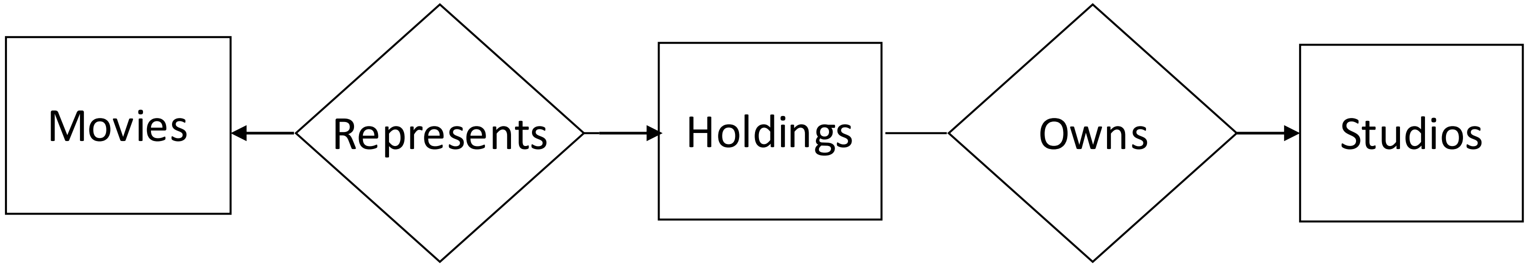
✓ **Conversion to SQL**

Design Principles

- Simplicity
- Avoiding redundancy
- Choosing the right elements and relationships
- Combining Relations

Simplicity

- Avoid introducing more elements into your design than those **absolutely necessary**



A poor design with an unnecessary entity set

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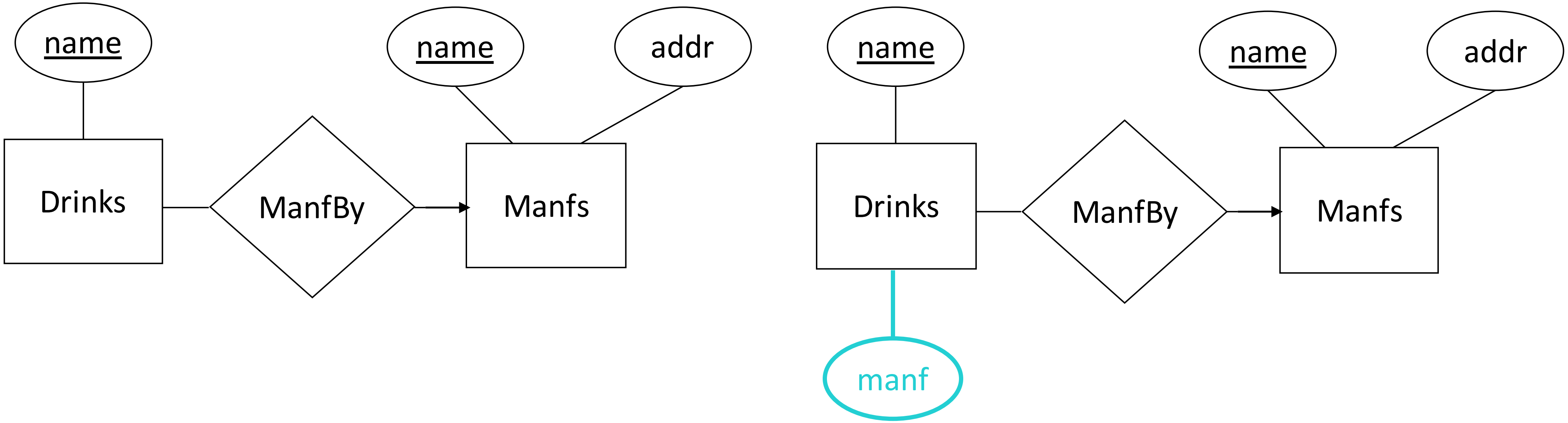
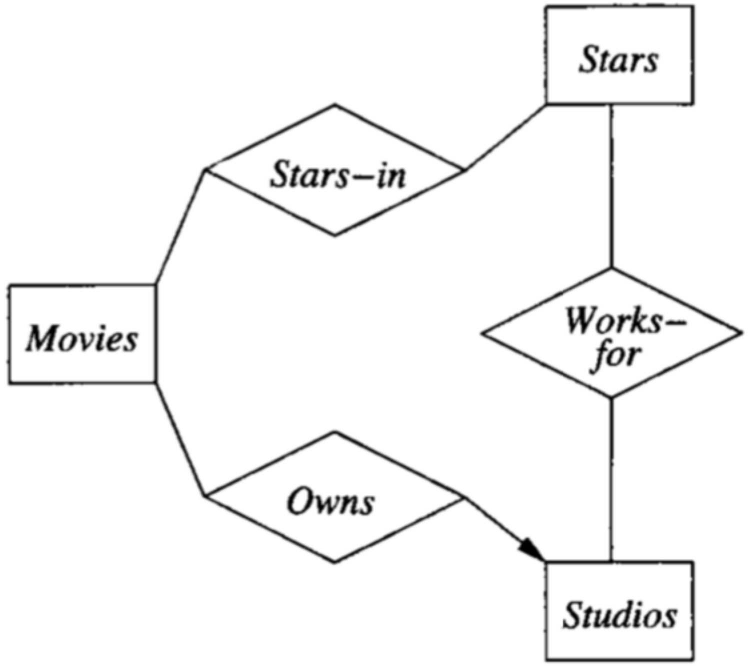
Avoiding Redundancy

- **Wastes space** and encourages **inconsistency**



Choosing Relationships

- Entity sets can be connected in various ways by relationships
- Adding to our design every possible relationship: not a good idea
  - Redundancy
  - Update anomalies
  - Deletion anomalies

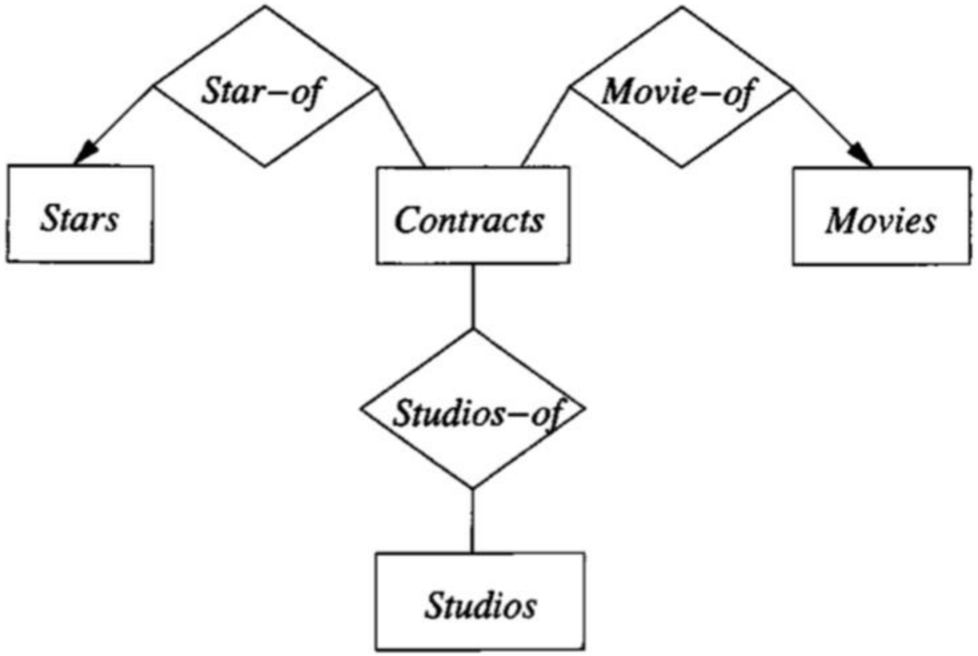


Adding a relationship between *Stars* and *Studios*

From Chapter 4, The complete book

Choosing Elements

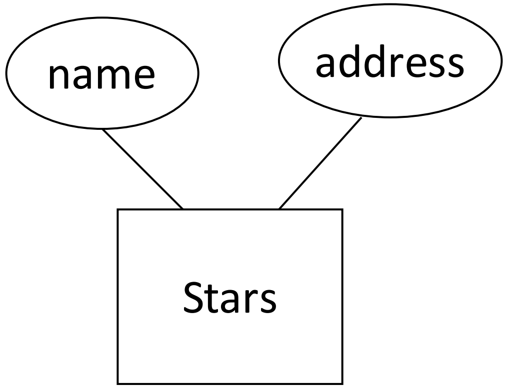
- Many choices are between using attributes and using entity set/relationship combinations.
- An attribute is simpler to implement than either an entity set or a relationship
- Cannot make everything into attributes!



From Chapter 4, The complete book

From Entity Sets to Relations

- Consider entity sets that are not weak
  - For each non-weak entity set, we shall create a relation of the same name and with the same set of attributes.
  - No indication of the relationships



name	address
Carrie Fisher	12 Maple St., Hollywood
Mark Hamill	456 Oak Rd., Brentwood
Harrison Ford	789 Palm Dr., Beverly Hills

From Chapter 4, The complete book

From E/R Relationships to Relations

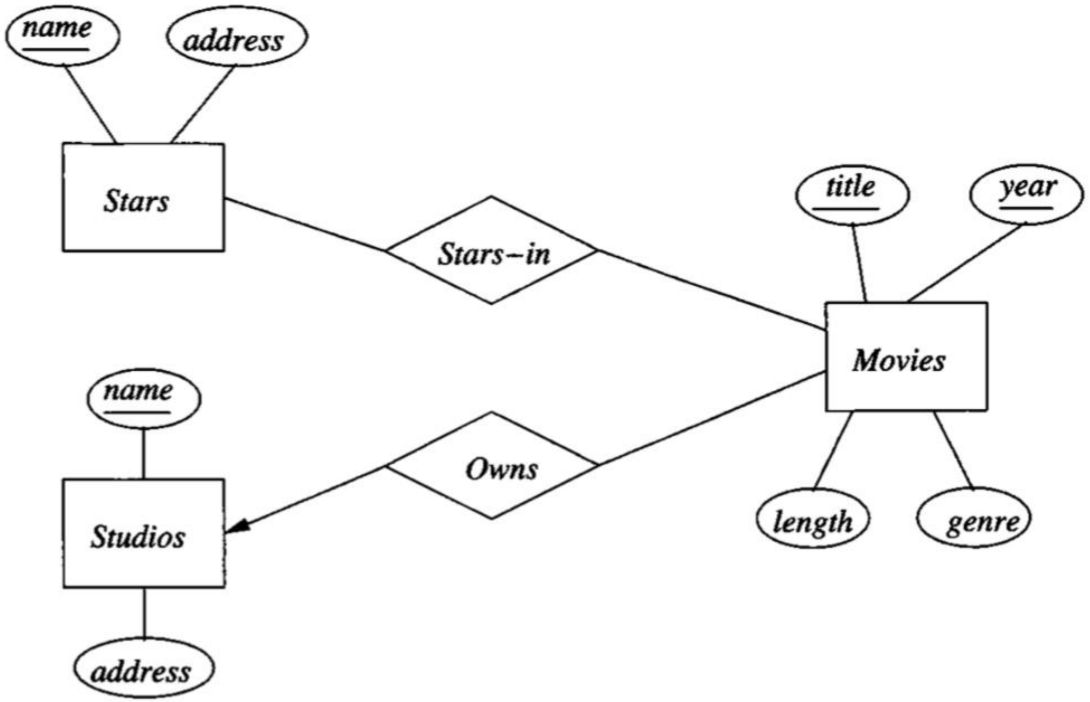
- Relationships in the E/R model are also represented by relations

- The relation for relationship *R*
  - Attributes
    - For each entity set involved in relationship *R:* its key attribute or attributes as part of the schema of the relation for *R.*
    - If the relationship has attributes: also attributes of relation *R.*
    - Entity set is involved several times in a relationship (in different roles): its key attributes each appear as many times as there are roles.

From E/R Relationships to Relations

Owns(title, year, studioName)

title	year	studioName
Star Wars	1977	Fox
Galaxy Quest	1999	DreamWorks
Wayne’s World	1992	Paramount



From Chapter 4, The complete book

Combining Relations

- All attributes of E
- The key attributes of F
- Any attributes belonging to relationship R

Movies:

title	year	length	genre
Star Wars	1977	124	sciFi
Galaxy Quest	1999	104	comedy
Wayne's World	1992	95	comedy

Owns:

title	year	studioName
Star Wars	1977	Fox
Galaxy Quest	1999	DreamWorks
Wayne's World	1992	Paramount

Combining relation Movies with relation Owns

title	year	length	genre	studioName
Star Wars	1977	124	sciFi	Fox
Galaxy Quest	1999	104	comedy	DreamWorks
Wayne's World	1992	95	comedy	Paramount

From Chapter 4, The complete book

Handling Weak Entity Sets

- The relation for weak entity set W
  - Include not only the attributes of W but also the key attributes of the supporting entity sets.
  - The supporting entity sets reached by supporting (double-diamond) relationships from W.

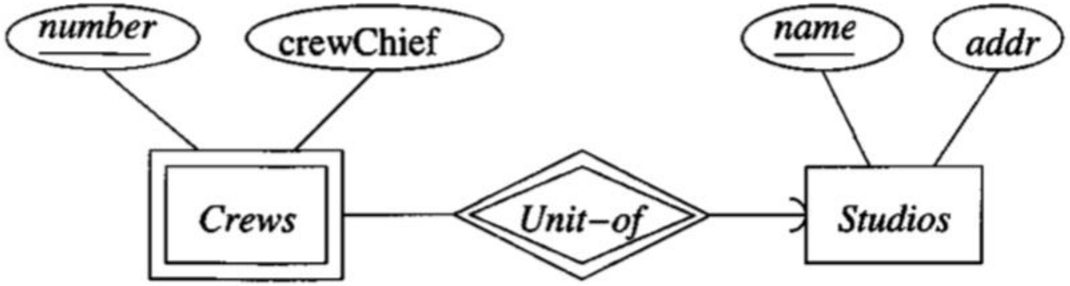
- The relation for any relationship in which the weak entity set W appears
  - Must use as a key for W all of its key attributes
    - Including those of other entity sets that contribute to W 's key.

- A supporting relationship R, from the W to a supporting entity set, need not be converted to a relation at all

Example: Handling Weak Entity Sets

```
Studios(name, addr)
Crews(number, studioName, crewChief)
```

~~Unit-of(number, studioName, name)~~



The crews example of a weak entity set

From Chapter 4, The complete book

Converting Subclass Structures to Relations

- Conversion Strategies
  - Follow the E/R viewpoint
    - Create a relation that includes the key attributes from the root and any attributes belonging to entity set
  - Treat entities as objects belonging to a single class
    - Example
      - Movies alone.
      - Movies and Cartoons only.
      - Movies and Murder-Mysteries only.
      - All three entity sets.
  - Use Null Values
    - Example

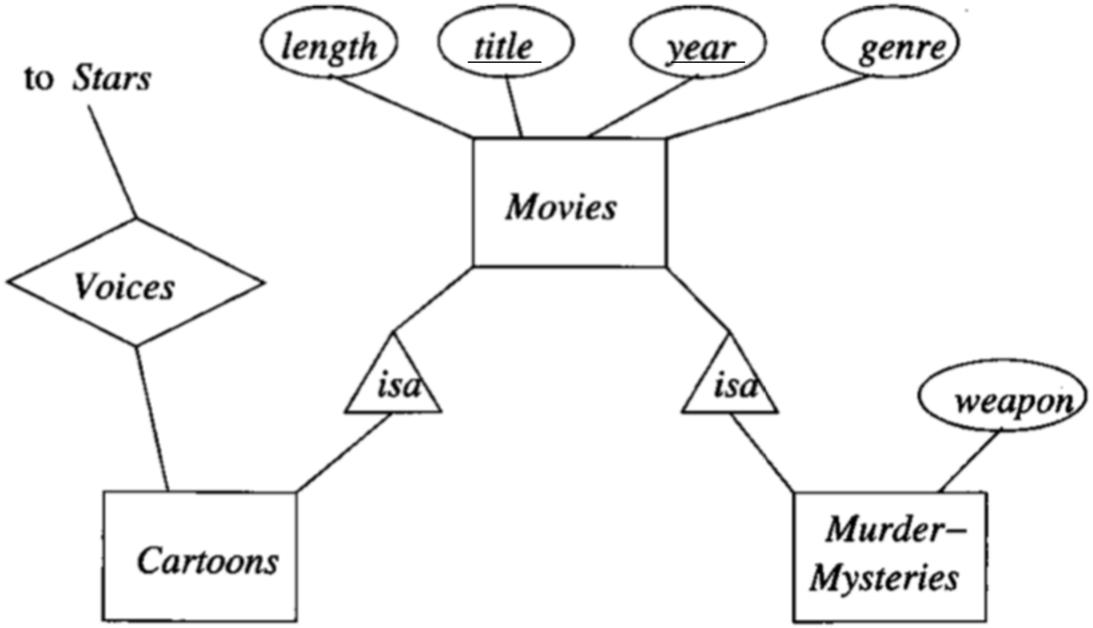
      `Movie(title, year, length, genre, weapon)`

      Movies that are not murder mysteries: NULL in the weapon component of their tuples

Example: Converting Subclass Structures

- E/R View

```
Movies(title , year, length, genre)
MurderMysteries(title, year, weapon)
Cartoons(title, year)
Voices(title, year, starName)
```



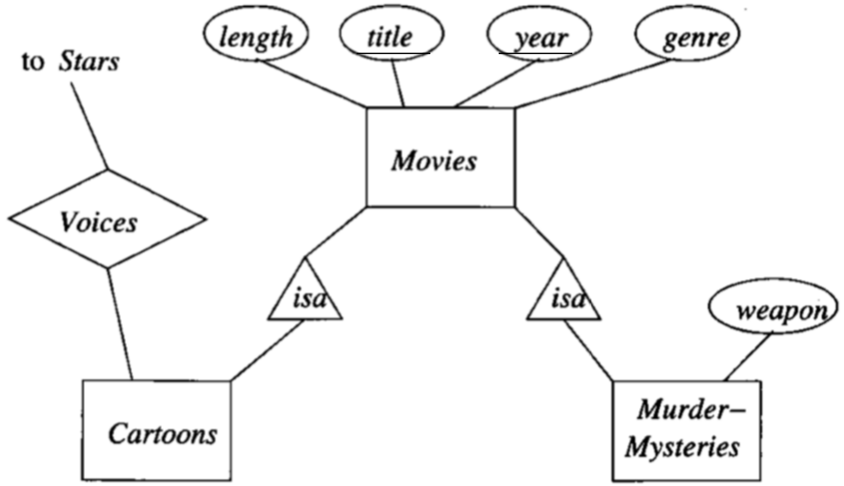
The movie hierarchy

From Chapter 4, The complete book

Example: Converting Subclass Structures

- Object-Oriented Approach

```
Movies(title, year, length, genre)
MoviesC(title, year, length, genre)
MoviesMM(title, year, length, genre, weapon)
MoviesCMM(title, year, length, genre, weapon)
Voices(title, year, starName)
```



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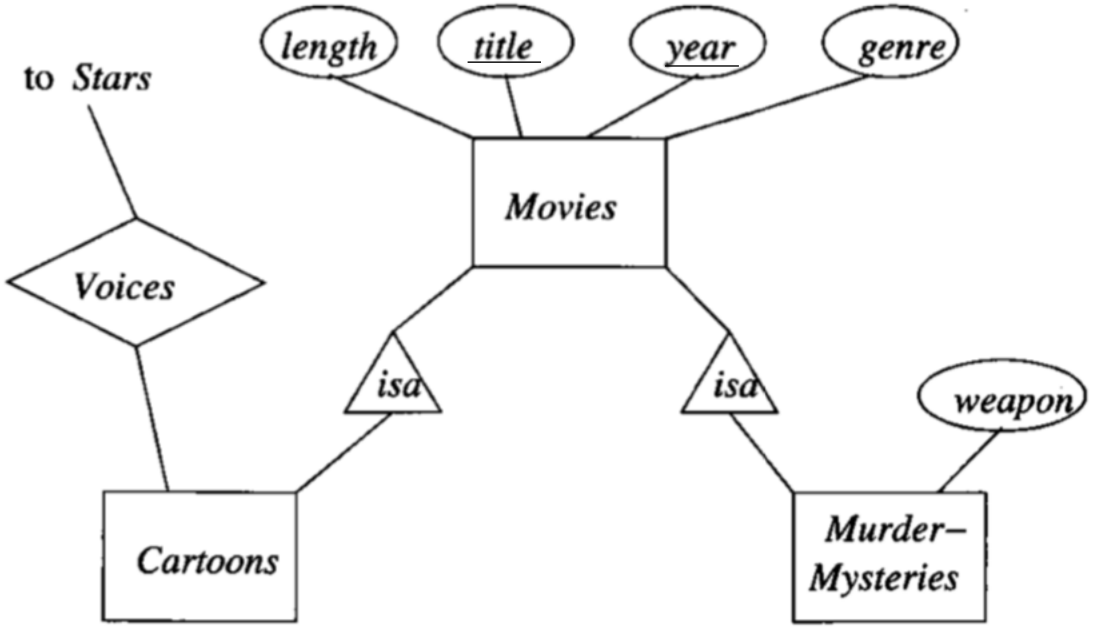
Example: Converting Subclass Structures

- Using Null Values

  `Movie(title, year, length, genre, weapon)`



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Example: Comparison

- We would like to reduce cost of queries
  - Expensive to answer queries involving several relations

- We would like not to use too many relations

- We would like to minimize space and avoid repeating information

Summary

- Database Design Process & E/R Model

- E/R Model:
  - Entity, Entity Set, Attributes, Relationships
  - Constraints, Multiplicity of Relationships, Multiway Relationships,
  - Subclasses, Weak Entity Sets

- From E/R to relations to schemas

Acknowledgements

[1] Database Systems: The Complete Book, 2nd Edition
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[3] CMPT 354 (Jiannan Wang - SFU)
https://sfu-db.github.io/cmpt354/

Additional references and resources used in preparation of this course are listed on
https://canvas.sfu.ca/courses/13083/pages/references
or mentioned on each slide.